SOLUTIONS

SE2228 ANAYSIS AND DESIGN OF ALGORITHMS
QUIZ #1 EXAM - SECTION I

ID: **Name:** **(60 min.)**
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1. Determine the value of the following sums:

a) $\sum_{i=10}^{i=60} i = \sum_{n=1}^{n=60} n - \sum_{n=1}^{n=9} = \frac{60 \cdot 61}{2} - \frac{9 \cdot 10}{2} = 1785$

b) $\sum_{i=1}^{i=55} i + 2 = \sum_{i=1}^{i=55} i + \sum_{i=1}^{i=55} 2 = \frac{55 \cdot 56}{2} + 2 \cdot (55) = 1650$

2.

a) Find the complexity of T(n) = $\log 26n^n$

$O(n) = n \log n$

b) Show that T(n) = $2n^3 + 5n^2$ is $\mathcal{O}(n^3)$

$2n^3 + 5n^2 <= c\,n^3$

$(c-2)n^3 >= 5n^2$

Take $c = 3$ then $\boxed{n = 5}$ is one possible answer

3. Order (in big O): 2^n, $n!$, $log(2^n)$, $n^3 + 99n^2$, $2^{log(n)}$, 3^n

$log(2^n)$, $2^{\log n}$, $n^3 + 99n^2$, 2^n, 3^n, $n!$

4. What is the solution of the following recurrence relation:

A(0) = 1 and A(n) = A(n-1) + 1

$A(1) = A(0) + 1 = 2$

$A(2) = A(1) + 1 = 3$

$A(3) = A(2) + 1 = 4$

$A(n) = n + 1$

5. What is the complexity of the following code segment? Explain your answer.

```
//All variables are already declared
for ( i= 0; i < n;i++) {

        temp=n; while(temp>0)

        temp=temp/2;

}
```

O(nlogn)

Outer loop is repeated n times and the inner while loop is executed logn times

6. If t1(n) is O(f1(n)) , t2(n) is O(f2(n)) and t3(n) is O(f3(n)) then

What is the complexity of t1(n)+t2(n)+t3(n) ?

Max of the three

7. What does the following function compute?

```
int abc (int x){

        if (x!=0) { return x + abc(x-1); }

        else { return x; }
```

Sum of the numbers 1 to X

8. Is recursion a good option if we have very large numbers of inputs and why?

No. Because a stack is created for storing local parameters and return addresses of recursive calls. Stack and its operations create overhead (Task time)